4.3.1.8 Antibiotics and painkillers

Content	Key opportunities for skills development
Students should be able to explain the use of antibiotics and other medicines in treating disease. Antibiotics, such as penicillin, are medicines that help to cure bacterial disease by killing infective bacteria inside the body. It is important that specific bacteria should be treated by specific antibiotics.	WS 1.4

Content	Key opportunities for skills development
The use of antibiotics has greatly reduced deaths from infectious bacterial diseases. However, the emergence of strains resistant to antibiotics is of great concern.	There are links with this content to Resistant bacteria (page 58).
Antibiotics cannot kill viral pathogens. Painkillers and other medicines are used to treat the symptoms of disease but do not kill pathogens. It is difficult to develop drugs that kill viruses without also damaging the body's tissues.	

Specification

Lesson 8 – Internal defences – white blood cells - Lymphocytes

Lesson 9 – Vaccination

Lesson 10 – Herd immunity upon reinfection

Lesson 11 – Medicinal drugs

Lesson 12 – Recreational and illegal drugs

Lesson 13 – End of Topic test

Lesson Sequence

Pupils should have been taught to:

Health

the effects of recreational drugs (including substance misuse) on behaviour, health and life processes

KS3 Links

From AQA GCSE Biology:

KS4: students will go on to learn about

Non-communicable and communicable diseases, internal and external defences, monoclonal antibody production and uses.

KS4 Links

Requisitions

1. **NONE**

2. **PRINT: Test**

Friday, 19 July 2024

Medicinal Drugs - Antibiotics & Symptom Relievers

Measles is a serious disease. A person can die from measles.

The table below shows the number of medically confirmed cases of measles in England and Wales between 2012 and 2015

Year	Number of medically confirmed cases of measles
2012	2030
2013	1843
2014	121
2015	91

(a) Suggest **one** reason why the actual number of cases of measles in England and Wales might be higher than is shown in the table above

(b) Calculate the percentage decrease in the number of medically confirmed cases of measles between 2012 and 2015

Keywords

Painkillers
Anti-inflammatories
Antibiotics

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Medicinal Drugs - Antibiotics & Symptom Relievers

Learning Objectives

- To define the term drug and medicinal drug
- To describe the terms antibiotic, painkillers, anti-inflammatories and give examples of each
- To describe the uses of these medicinal drugs.
- To explain why antibiotics cannot be used on viral infections

Keywords

Painkillers
Anti-inflammatori es
Antibiotics

What is a drug?

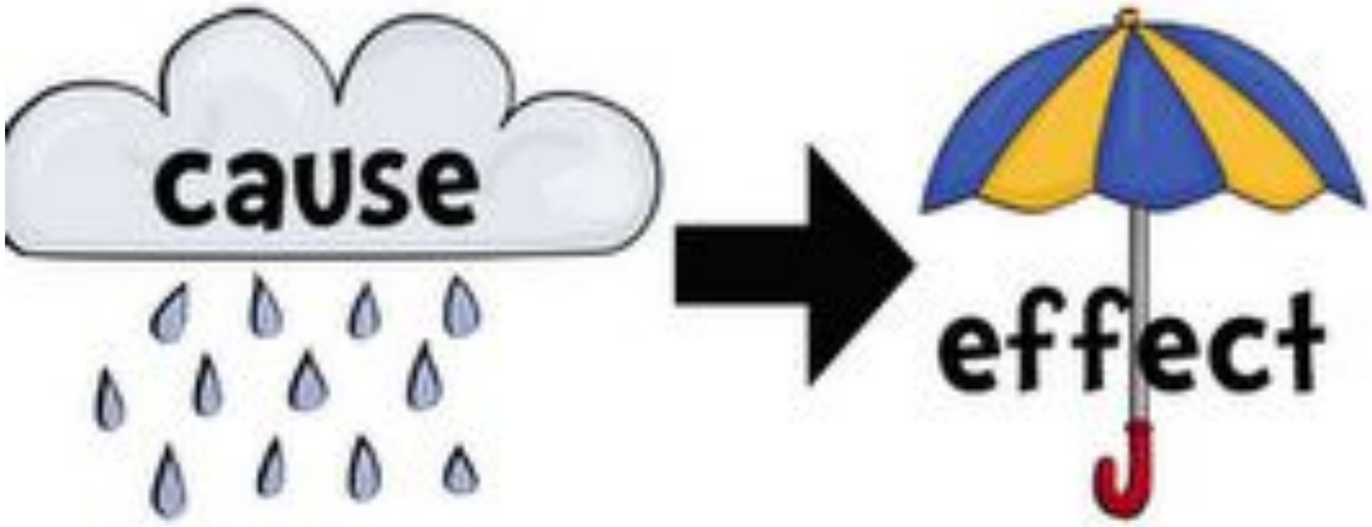
Drug

A medicine or other substance which has a physiological effect when ingested or otherwise introduced into the body.

Medicinal drug

A drug or other preparation for the treatment or prevention of disease.

Why take drugs?



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Types of drugs

Painkillers **Anti-inflammatories** **Antibiotics**

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Why take paracetamol and ibuprofen?

Paracetamol:

- Pain reliever

Ibuprofen:

- Pain reliever

Painkillers and anti-inflammatories are chemicals that relieve the symptoms but do not kill the pathogens

Cause or symptom? – sort the facts into the table

Bacteria releases toxins damaging cells

Body increases temperature to inactivate virus

Congested nose – causing difficulty in breathing

Cough

Muscle pain

Virus causes host cells to die

Overgrowth of bacteria damages cells

Headache

Cause	Symptom

Correct in GREEN PEN

Bacteria releases toxins damaging cells

Body increases temperature to inactivate virus

Congested nose – causing difficulty in breathing

Cough

Muscle pain

Virus causes host cells to die

Overgrowth of bacteria damages cells

Headache

Cause	Symptom

When you are injured, sensory nerve endings send pain messages to your brain.

Painkillers stop these nerve impulses, so you feel little or no pain.

Where do medicines come from?

Many painkillers are based on two natural drugs:

- **aspirin**, from willow bark
- **opiates**, from poppies

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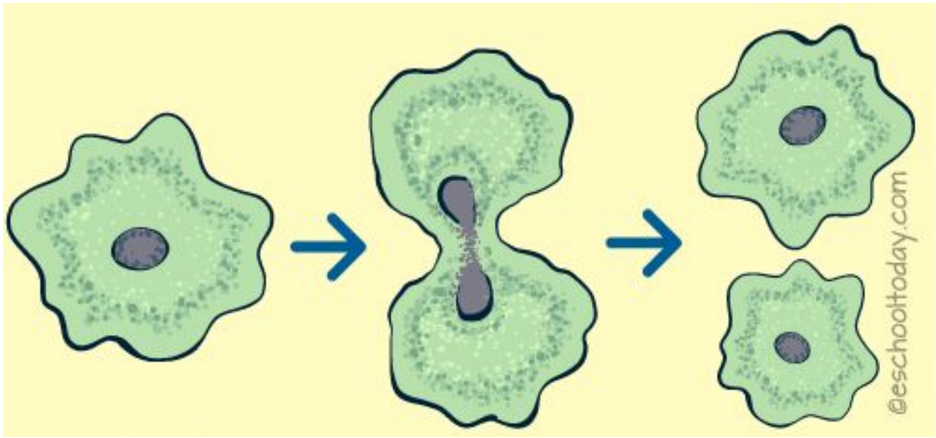
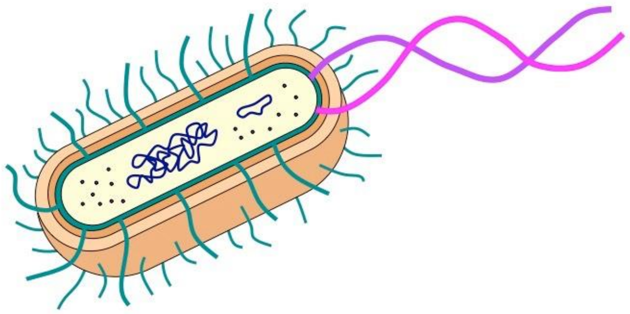
ACTIVATION

What are antibiotics?





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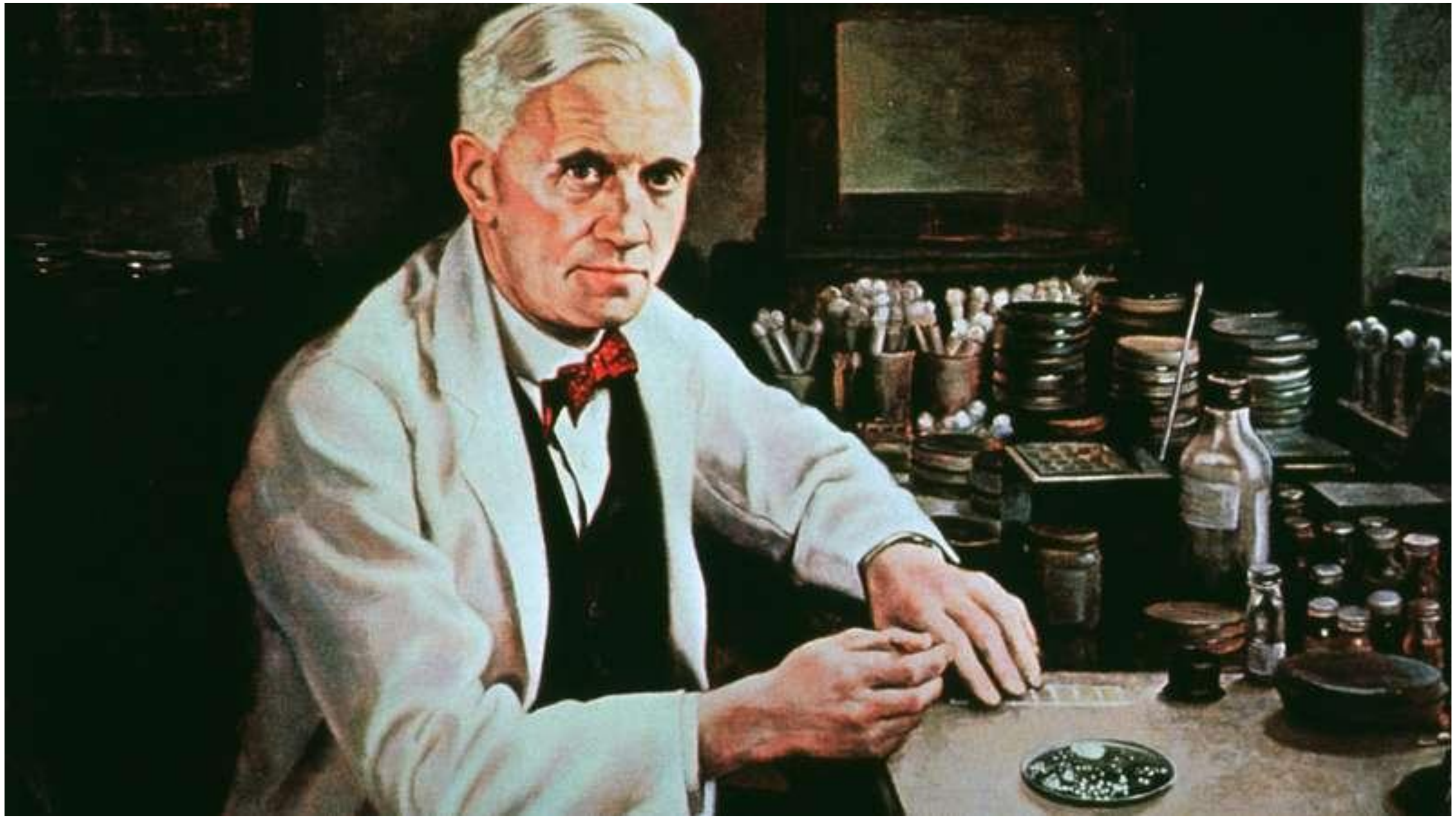
The accident that changed medicine



Answer the following questions while watching the video

1. What was Alexander Fleming investigating?
2. What unexpected finding did Fleming find around the mould?
3. What did Fleming name the chemical released by the mould?
4. How did the discovered chemical destroy bacteria?
5. Why is the chemical not harmful to human cells?

Do Now | I Do | We Do | You Do | Test

Correct using green pen

1. What was Alexander Fleming investigating?

Properties of infectious bacteria

2. What unexpected e mould?

A region of no bacteria

3. What did Fleming e mould?

Penicillin

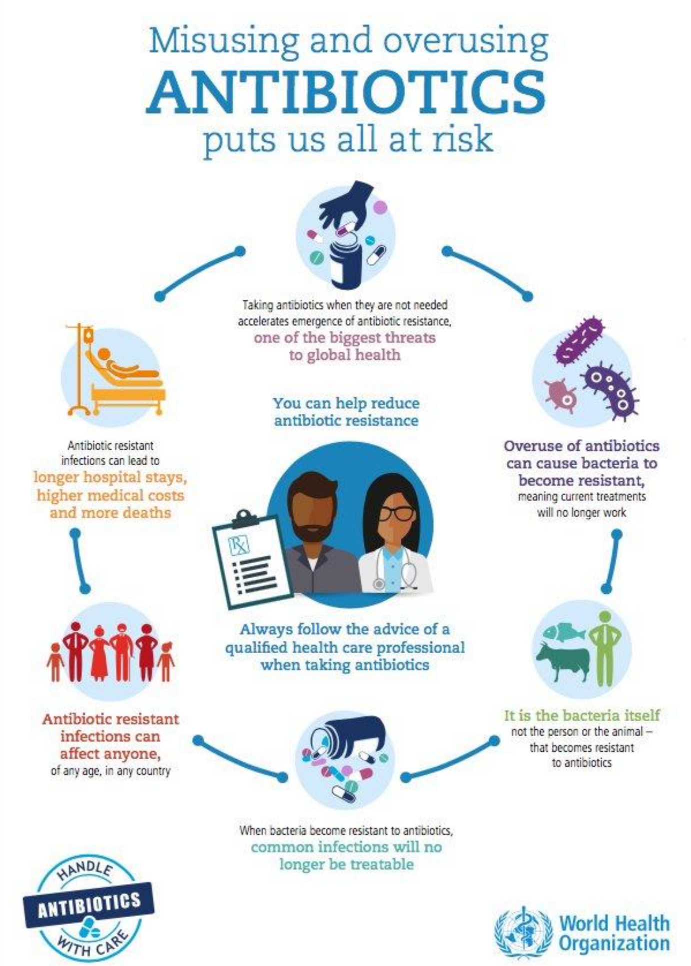
4. How did the disco

- Disrupts synthesis of cell wall
- Releases highly reactive molecule which causes additional damage

5

Because human cells do not have a cell wall

Using antibiotics.



Do Now | I Do | We Do | You Do | Test

Respond to the following NHS complaints

Dear NHS complaints,
I am appalled by the incompetence of your staff. Today I went to my GP as my son is showing signs of a flu, however the GP did not prescribe him antibiotics. Why is this?

Karen

Dear NHS complaints,
I was involved in a car accident, where I injured my arm. Should I be given antibiotics to prevent infection?

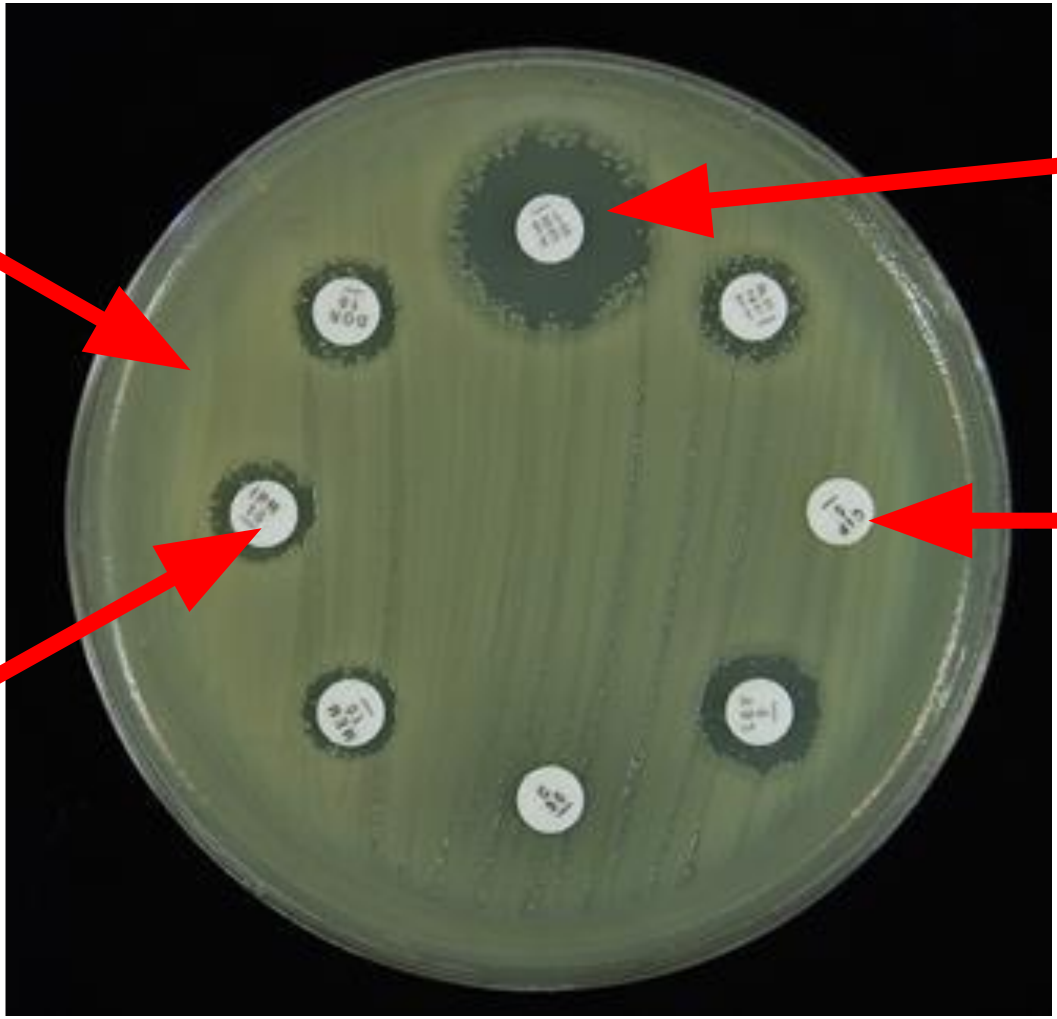
Jasmine

Do Now | I Do | We Do | You Do | Test

Which antibiotic is the most effective?

Bacteria

Bacteria free zone



Control

Paper disk soaked in water

Paper disk soaked in antibiotic

Answer the following questions in your books.

1. Which disk contains the most effective antibiotic? Give a reason for your answer . [2]

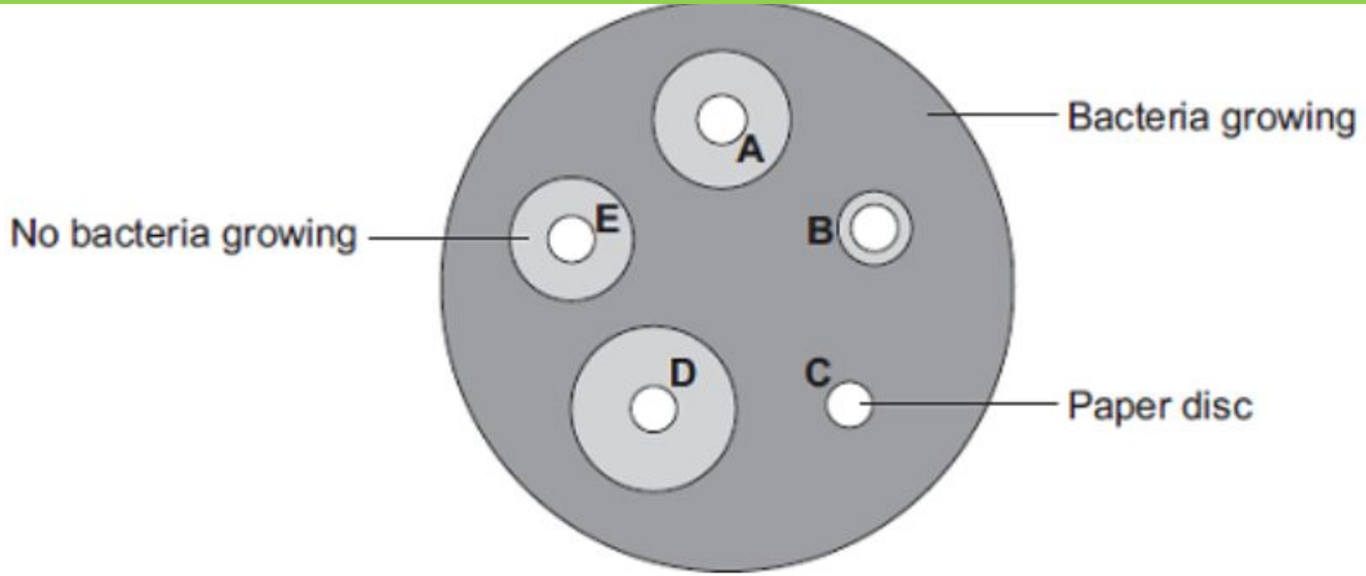
D – has a the greatest area with no bacteria growing

2. Which disk is t

C – does not have a region without bacteria

3. Why

To show that the paper disk does not have antibacterial properties



Stretch

Measure the diameters of the bacteria-less region

Do Now | I Do | We Do | You Do | Test

Correct these false statements

1. Painkillers **do not** help to kill bacteria

Painkillers **are helpful** to relieve symptoms caused by all infections including viral infections

3. Antibi

Antibiotics assist in **bacterial** infections

Antibiotics work by breaking down cell walls and **preventing** cell reproduction

Antibiotics are specific. One antibiotic **cannot** be used to treat all infections.

An **antibiotic** is a drug that helps to cure a bacterial disease by killing the infective bacteria **inside** the body.

Different bacterial infections need a **different** antibiotic.

Antibiotics **cannot** be used to **treat viral** pathogens.

Penicillin is a well known antibiotic medicine.

Using antibiotics has greatly **reduced** deaths.

It is difficult to develop **drugs** to kill **viruses** without harming body tissues because viruses live and reproduce inside cells.

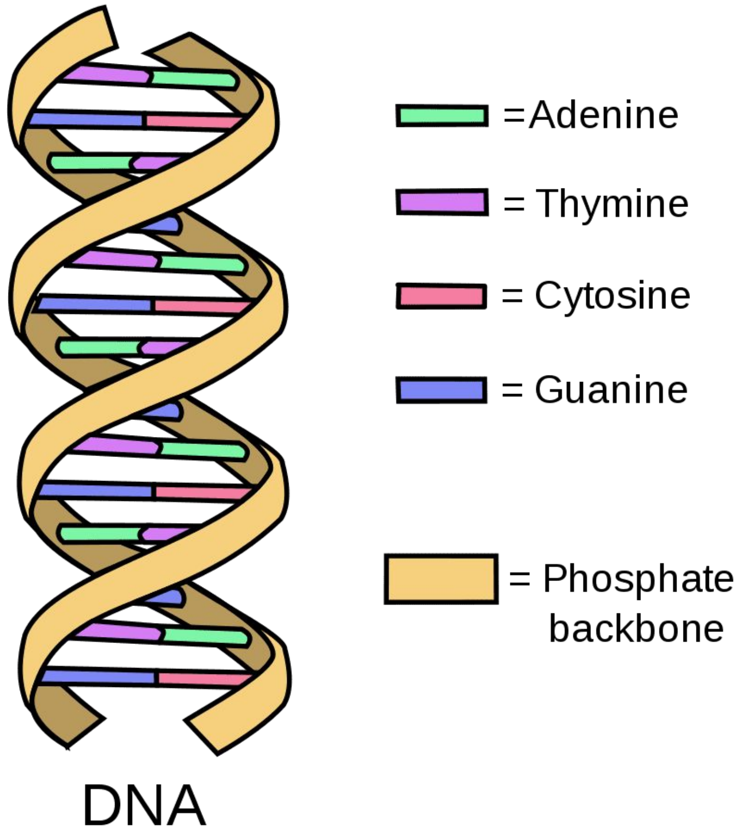
School

ACTIVATION

What is a mutation?

- Change in the sequence of bases – coding of the DNA
- This can be by mistake!

- Can cause advantageous or harmful changes to the organism.



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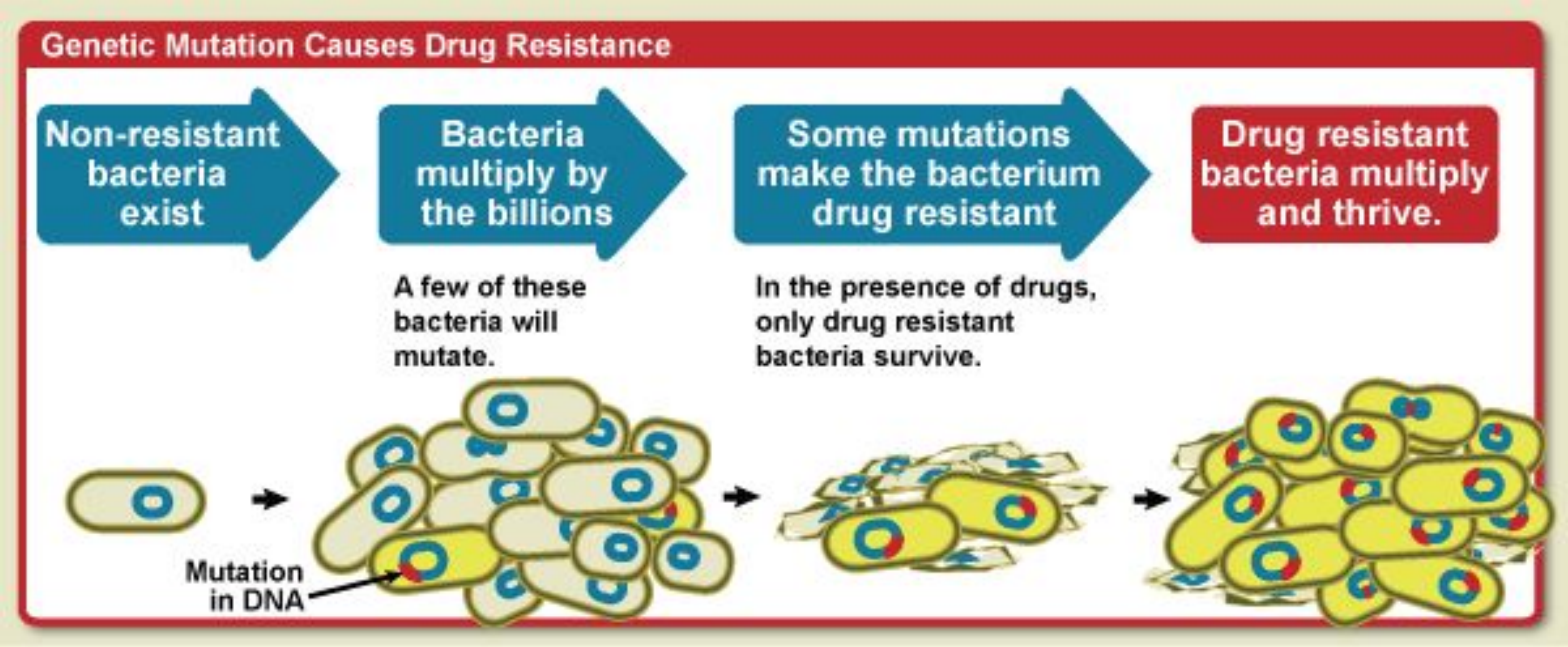
DO NOW

Antibacterial resistance



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ACTIVATION

What leads to further antibiotic resistance?

- Incomplete courses of antibiotics
- Overprescribing antibiotics
- Prescribing antibiotics for viral infections

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ACTIVATION

Why can't we use antibiotics for viruses?

They don't have cell walls

Live inside of cells- difficult to reach

Antiviral drugs

Specific to each virus, slow down the development of viruses however cannot completely eradicate them without damaging tissues.

1. Name and describe 3 ways the human body defends against the entry of pathogens

Skin – barrier, nose - nasal hairs , mucus and cilia, trachea & bronchi – mucus to catch, cilia to remove from lungs, stomach - HCl acid kills.

2. Which one of these statements is true?

Painkillers are used to treat the symptoms of a disease and kill the pathogens which cause it.

Painkillers are used to kill the pathogens causing a disease.

Painkillers are used to treat the symptoms of a disease but do not kill the pathogens.

Painkillers are used to treat the symptoms of a disease but do not kill the pathogens.

3. What is the definition of a pathogen?

Micro-organisms which cause infectious disease in animals & plants

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CONSOLIDATION

Describe how bacteria gains resistance

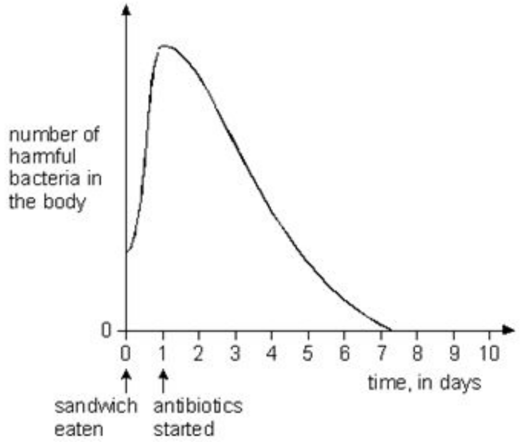
- Bacteria is subjected to a selective pressure e.g. antibiotics
- Some individuals acquire an advantageous mutation
- This allows them to survive
- These reproduce
- The whole colony is now antibiotic resistant

A quick celebration of learning (i.e: TEST!)

Q1.

One evening Jenny and Leah ate chicken sandwiches which had been in their school bags all day. There were harmful bacteria in the food. The next day both girls became very ill. Their doctor gave them antibiotics to take for eight days.

The graph represents how antibiotics affect the number of bacteria in the body.



(a) Use the graph to explain why the girls did **not** become ill until the day after eating the sandwiches.

....

....

1 mark

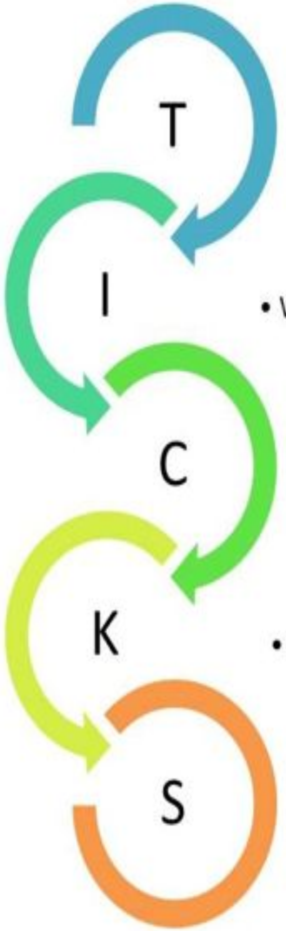
(b) After taking the antibiotics for eight days Jenny was completely better. Explain why she got better.

....

....

1 mark

(c) Leah should have taken the antibiotics for eight days. She felt much better after five days and stopped taking the antibiotics. Two days later she felt very ill again. Use the graph to help you explain why Leah became ill again.



- **Topic?**
- What is the context of the question?
- **Information?**
- What relevant information are you given in the question?
- **Command?**
- What are you being asked to do?
- *e.g. state, describe, explain*
- **Knowledge?**
- What do you know that can help you answer the question?
- **Satisfied?**
- Have you satistfied the examiner? 5 marks = 5 points

Do Now | I Do | We Do | You Do | Test

Q1.

(a) any **one** from

- there were not enough bacteria in the food **or** body
  accept 'the bacteria had to grow first'
- the bacteria multiplied by the next day

1 (L6)

(b) the antibiotic **or** medicine killed all the bacteria
accept 'the antibiotics got rid of all the bacteria'
***or** 'there were no bacteria left'*

1 (L5)

(c) any **one** from

- antibiotic **or** medicine had not killed all the bacteria
  accept 'not all the bacteria had gone'
- there were still bacteria left alive

1 (L6)

- the bacteria multiplied
  accept 'the population rose again'
  accept 'they could grow again'
  accept 'they reproduced again'

1 (L6)

(d) any **one** from

- it slows down reproduction

  accept 'it stops them reproducing'

  or *'it stops them breeding'* ***or*** *'it stops them multiplying'*

- it is too cold for the bacteria to divide **or** reproduce

  accept 'it stops them growing'

  accept 'slows down growth'

  do ***not*** *accept 'they are dormant'*

  do ***not*** *accept 'it freezes them'*

A quick celebration of learning (i.e: TEST!)

Q1.

Antibiotics are used to kill some types of pathogen.

(a) Which illness could be treated with an antibiotic?

Tick **one** box.

AIDS ☐

Measles ☐

Salmonella ☐

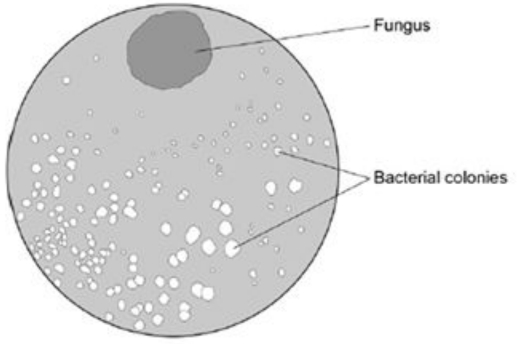
Type 2 diabetes ☐

(1)

Alexander Fleming discovered the antibiotic penicillin.

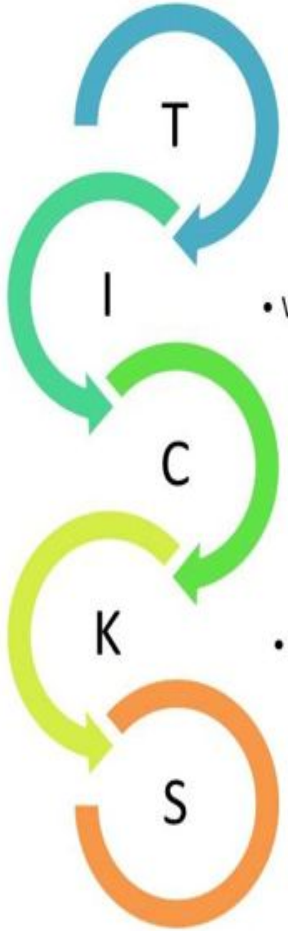
He noticed that one of his Petri dishes containing bacteria had become contaminated with a fungus.

The diagram shows the Petri dish.



(b) Read the information about the discovery of penicillin.

Draw **one** line from each piece of information to its description.



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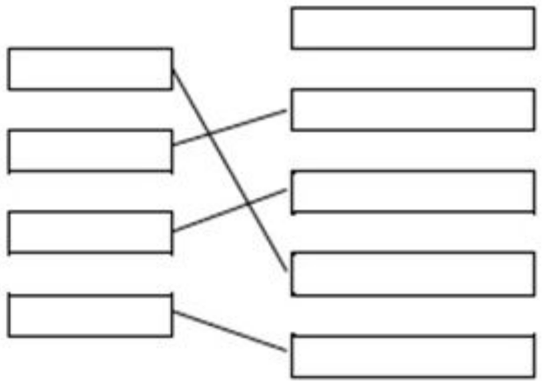
Q1.

(a) salmonella 1

(b)



1
1
1
1

(c) lower concentration of antibiotic / chemical further from the fungus

allow less antibiotic / chemical further from the fungus

1

(d) lead to mass production of antibiotics
or
lead to development of other antibiotics

1

reduced infection by bacteria
or
antibiotics have saved many lives

1

[8]

Do Now

Test